3.3.1 Introduction to Differentiation

In sections 3.1 & 3.2 we found the derivatives of functions using *the definition of the derivative.* We noticed that this process was often very complicated a required a lot of work. Our goal in the following sections is to use our knowledge of *the definition of the derivative* to develop shortcuts in the process of finding the derivatives of various functions.

Recall:

Definition: The **derivative of a function *f***, denoted by $f'(x)$ is,

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

if this limit exists.

Example 1: In parts (a) through (c) find $\frac{dy}{dx}$ using the definition of the derivative.

a) $y = x$
b) $y = x^2$
c) $y = x^3$
d) Based upon your answers above, what would you guess is the first derivative of $f(x) = x^4$? What about $g(x) = x^{97}$?

The next theorem will confirm that our guess in part (d) is correct, and it is of great importance.

Theorem (The Power Rule): If n is any real number, then $\frac{d}{dx}\left(x^n\right) = nx^{n-1}$.

Note: There is a proof of this in the textbook when n is a natural number which uses the Binomial Theorem. We will need implicit differentiation, which we will learn about later, in order to give a complete proof.

Example 2: Compute each of the following derivatives.

a) $\frac{d}{dx}\left(\frac{1}{x}\right)$

b) $\frac{d}{dt}\left(t^{1.7}\right)$

c) $\frac{d}{ds}\left(\sqrt[5]{s^7}\right)$

Question: Could we use the power rule to figure out: $\frac{d}{dx}\left(x^x\right)$*?*

Example 3: Consider the curve given by $f(x) = x\sqrt{x}$.

a) Find the derivative of y with respect to x.

b) Find the slope of the of the tangent line to the curve at $(4,8)$. Then find the equation of the tangent line to the curve at $(4,8)$.

NOTE: _Simplifying_ before trying to take a derivative is always a good idea.

We now present some fairly intuitive theorems which will help us differentiate a larger variety of functions.

Theorem: Suppose that c is a constant and f and g are differentiable functions. Then the following rules hold.

1. **The Constant Rule:** $\frac{d}{dx}(c) = 0$
2. **The Constant Multiple Rule:** $\frac{d}{dx}\left[c \cdot f(x)\right] = c \cdot \frac{d}{dx}\left[f(x)\right]$
3. **The Sum & Difference Rules:** $\frac{d}{dx}\left[f(x) \pm g(x)\right] = \frac{d}{dx}\left[f(x)\right] \pm \frac{d}{dx}\left[g(x)\right]$

Proof (of 1):

Proof (of 2):

Note: The proof of 3 (specifically the Sum Rule) will be an assigned problem for Homework Assignment 4.

Example 4: Compute $\frac{d}{dx}\left(x^7 - 22x^3 + 4x^2 - 6x + \sqrt{5}\right)$

Note: Normally, we will not show the application of each rule. In fact, as you get better at applying the rules of differentiation, you should be able to go right to the final answer in a problem like the example above.

Example 5: Compute $\frac{d}{dx}\left[\frac{\sqrt{x}-2}{\sqrt{x}}\right]$

Exploration:
We would like to compute the derivative of the exponential function $f(x)=b^x$ where b is some positive real number constant.

Definition (The number *e*): The number e is the real number such that

Note: One may recall from Precalculus that $e \approx 2.718281828459$

Note: The exploration and definition immediately imply the following theorem:

Theorem:

__Example 6:__ Suppose that $f(x) = -2x + e^x$.

a) Find $f'(x)$.

b) At which point(s) on the curve determined by *f* is there a horizontal tangent line?